KEMENTERIAN PENDIDIKAN TINGGI

PROGRAM PENINGKATAN SKIL USAHAWAN DIGITAL



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ASCEND 2030

Handling Digital Evidence at the Scene



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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 2.0

Handling Digital Evidence at the Scene

Lesson overview: recognise, protect and record evidence



LESSON 2.0 · OVERVIEW

Evidence at the Scene

What beginners are most likely to do first

BY THE END YOU WILL BE ABLE TO

Identify likely digital evidence

Including accounting records kept by a business

Explain why rules of evidence matter

And how they shape the way evidence is handled

Prepare and secure a scene

So that nothing is changed before it is recorded

Bag, tag and record custody

Fill in an evidence form and a custody log

LESSON 2.0 · ROUTE

Three Short Sub-Lessons

One idea and one desk activity in each

LESSON 2.1

Identifying Digital Evidence

What counts as digital evidence, and the rules that decide its use.

ACTIVITY **Evidence list**

LESSON 2.2

Preparing and Securing the Scene

Planning a search, and keeping the scene unchanged.

ACTIVITY **Laptop left on**

LESSON 2.3

Bagging, Tagging and Custody

Labelling, sealing, moving and storing evidence.

ACTIVITY **Custody log**

Practise only on training devices and forms. Never handle a real device or real evidence in this course.

LESSON 2.0 · THE CASE

The Teratai Case So Far

From Lesson 1 to the accounts office

SO FAR

Payments were made for materials never delivered, approved by the accounts payable executive.

NOW

Investigators go to the accounts office to find, protect and record the evidence.

ALREADY KNOWN

- **The executive's laptop**
- **A USB drive in the desk**
- **A mobile phone**
- **More to find: your list in 2.1**

The question for this lesson: what must be found, protected and recorded before anything is examined?

UP NEXT

Identifying Digital Evidence

We start with what digital evidence is, and the rules that decide whether a court will accept it.

- **What digital evidence is**
- **Rules of evidence**
- **Evidence in a financial case**

DFBOOK DIGITAL FORENSICS · LESSON 2.1

Identifying Digital Evidence

What counts as digital evidence, and the rules for using it



IN THIS LESSON

1. What digital evidence is
2. Rules of evidence
3. Evidence in a financial case

LESSON 2.1 · OVERVIEW

Identifying Digital Evidence

Digital evidence is everywhere in a modern office, but a court will only accept it if it is handled by the rules. This lesson covers both.

BY THE END YOU WILL BE ABLE TO

1. **Identify likely sources of digital evidence**
2. **Tell generated from stored records**
3. **Explain why rules of evidence shape handling**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Case study: one file's hidden details identified a killer



A floppy disk

Sent to a TV station in 2005

File details

A church name and the user “Dennis”

An arrest

Dennis Rader, the BTK killer, confessed

Metadata: data about a file, such as its author, its organisation and its dates, stored with the file.

Digital evidence can point to a person, but only if it is found, kept and presented in a way a court accepts.

INTRODUCTION · ROADMAP

Your Route Through Lesson 2.1

Each part answers one question, in order

1

PART 1

What digital evidence is

What counts as digital evidence, and how is it gathered?

2

PART 2

Rules of evidence

What decides whether a court will accept it?

3

PART 3

Evidence in a financial case

How do these rules apply to business records?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

What Digital Evidence Is

A working definition, and the standard tasks around it

01

IN THIS PART YOU WILL LEARN TO

- Define digital evidence
- Name the four general tasks with evidence

KEY TERMS

Digital evidence **ISO/IEC 27037** **SWGDE**

PART 1 · DEFINITION

Defining Digital Evidence

One sentence, and what each part of it means

Digital evidence is (1) anything stored or sent in digital form. US courts treat it as (2) physical evidence, handled to (3) agreed standards.

1 Stored or sent

On a drive, a phone or a server, or moving across a network.

2 Physical evidence

Treated in US courts like a weapon or a paper document. For Malaysia, see Part 2.

3 Agreed standards

ISO/IEC 27037 and the Scientific Working Group on Digital Evidence (SWGDE).

ⓘ Each country decides what its courts accept. Some once required every item to be printed.

PART 1 · CATEGORIES

Where Digital Evidence Is Found

Devices that may need to be seized, depending on the case

- **Computers and laptops**
- **Smartphones and tablets**
- **USB devices**
- **CDs and DVDs**
- **Printers and scanners**
- **Cameras**
- **Smart TVs and game consoles**
- **Network devices**

Also off the premises: cloud storage and remote servers. In a criminal case, whole drives are seized; in a company case, a few specific items may be enough.

PART 1 · METHOD

Working with Digital Evidence

Four general tasks, and what each one achieves

THE TASK	WHAT IT ACHIEVES
Identify Find the information or artifacts that might hold evidence.	FIND IT
Collect, preserve and document Take it without changing it, and record what was done.	PROTECT IT
Analyse, identify and organise Work out what the evidence shows, and arrange it.	UNDERSTAND IT
Rebuild or repeat Repeat the steps to show the results can be reproduced.	PROVE IT

PART 1 · STRUCTURE

Collecting Evidence Systematically

Four rules that reduce confusion, loss and damage at a scene

One team collects

Only one team collects and catalogues evidence, where practical.

Same procedures

If several teams are needed, all follow the same procedures.

One lead examiner

A lead examiner controls collecting and cataloguing.

Standard forms

The same forms track every item (Lesson 2.3).

Across borders: when raids happen in several countries at once, applying the highest standard everywhere lets the evidence be used in any of them.

PART 2 OF 3

Rules of Evidence

What decides whether a court will accept digital evidence

02

IN THIS PART YOU WILL LEARN TO

- Explain hearsay and the business-record exception
- Tell generated from stored records

KEY TERMS

Hearsay **Business record** **Best evidence**

PART 2 · CONCEPT

Why Digital Evidence Needs Care

Three reasons the same strict controls apply to every case

Cases change

Evidence admitted in a criminal case may be used in a civil suit, and the other way round.

Easily changed

Digital data can be altered, and a change shows only when the original is compared with a copy.

Copies look identical

A copy is hard to tell from the original, so courts need proof of how it was made.

Rules of evidence: the laws that decide what a court may accept. US rules differ from Malaysia's, which are on a later slide.

PART 2 · DEFINITION

Hearsay and Business Records

A general rule, and the exception that matters most for financial records

Hearsay: a statement repeated in court by someone who did not witness the event, such as an office rumour. It is generally not accepted.

THE BUSINESS-RECORD EXCEPTION · A RECORD CAN BE ACCEPTED IF IT WAS

1. **Made at or near the time**
   Recorded when the activity happened.
2. **By someone with knowledge**
   Or from information they supplied.
3. **Kept as regular practice**
   Part of the normal course of business.



Example: a bookkeeper's entries for expenses, profits, losses and payroll, in a ledger book or in accounting software.

PART 2 · COMPARISON

Computer-Generated and Computer-Stored

Two kinds of digital record, compared on the same four points

	Computer-generated	Computer-stored
Created by	A computer process, not a person	A person, saving their work
Examples	System logs, proxy server logs	Spreadsheets, documents, bookkeeping
To be trusted	Show the program works correctly	Show who made it, and it was not altered
Hearsay?	Usually not	Yes: needs the business-record exception

Some records are both: a spreadsheet's formulas run on figures a person typed.

PART 2 · COMPARISON

Originals and Copies

The best evidence rule, and when a copy will do

THE BEST EVIDENCE RULE

The original is preferred

To prove what a document says, the original is ordinarily required. Under US rules, an accurate printout counts as an original.

WHEN A COPY WILL DO

A correct copy is accepted

A bit-stream copy (an exact bit-by-bit copy) made and kept correctly can be admitted, unless the original is challenged.

Original lost?

The first good copy becomes secondary evidence; the lawyer explains why.

Server in use?

Use the least disruptive way to collect from it.

PART 2 · LEGAL POINTERS

Rules of Evidence in Malaysia

Where to look in the Evidence Act 1950, instead of US rules

Section 3

The meaning of “document” includes electronic records.

Section 62, Explanation 3

A document produced by a computer is primary evidence.

Sections 90A to 90C

When computer-produced documents are admissible, including a certificate under section 90A(2).

Before relying on these: confirm the current wording with a legal adviser. They come from the UNODC electronic-evidence summary for Malaysia (December 2022).

PART 3 OF 3

Evidence in a Financial Case

Business records, authorship, and a fraud's evidence lessons

03

IN THIS PART YOU WILL LEARN TO

- Explain how authorship is proved
- Sort records by how they were made

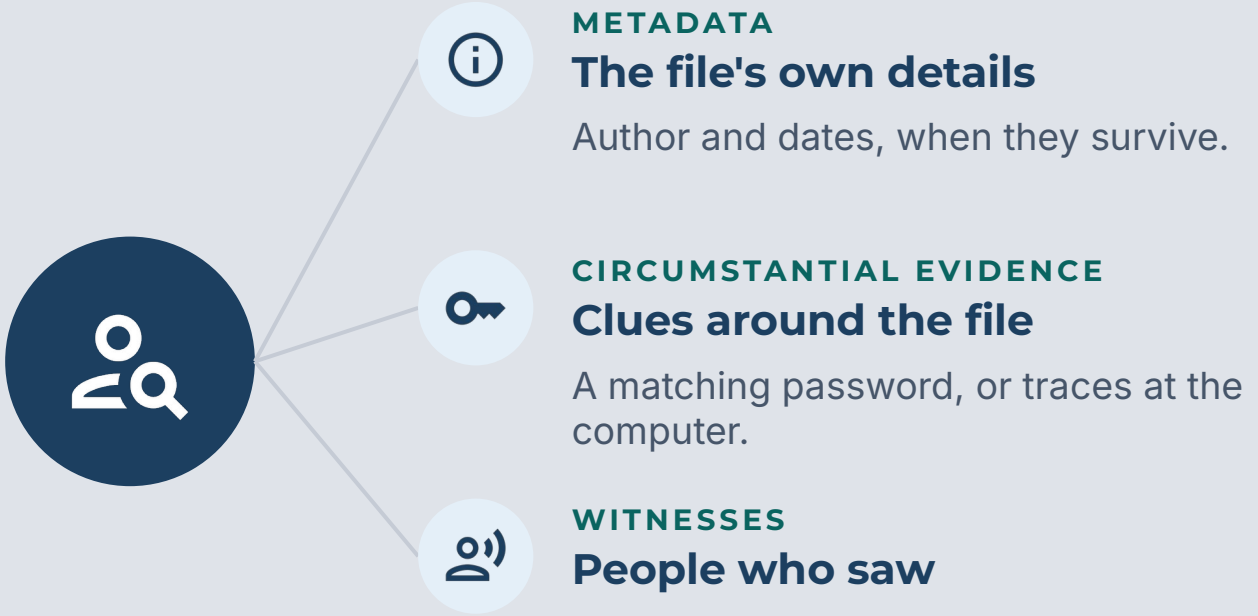
KEY TERMS

Circumstantial evidence **Personal data**

PART 3 · STRUCTURE

Proving Who Made a Record

Three kinds of clue, and a case where none was found



METADATA

The file's own details

Author and dates, when they survive.

CIRCUMSTANTIAL EVIDENCE

Clues around the file

A matching password, or traces at the computer.

WITNESSES

People who saw

Someone saw the suspect at the computer then.

WHY IT MATTERS

Files recovered from free space (space no file is using) usually lose their metadata, so the author is not recorded.

A CASE STUDY

A camera was proved to have made the suspect's videos, but not who used it. With no other clues, the lawyer did not use the evidence.

PART 3 · AWARENESS

The Enron Collapse

Case study: a fraud that collapsed in 2001, and what its evidence taught

20,000+

Jobs lost

When the energy company collapsed.

US$3 billion+

Retirement savings lost

By employees and retirees.

21

Executives charged

Including fraud and money laundering.

⚠ **Lesson for evidence handling:** Enron's emails were released with staff's personal data (details that identify a person) still in them; lawsuits followed. Its auditor collapsed after shredding documents.

PART 3 · WORKED EXAMPLE

Sorting Records by How They Were Made

Six records from an accounts office, sorted into three groups

THE CASE

Which records were made by a computer, by a person, or by both?

COMPUTER-GENERATED	COMPUTER-STORED	BOTH
Logon records	**Invoice document**	**Payment spreadsheet**
Proxy server log	**Email message text**	**Ledger with totals**
Show the program works.	Show who made it.	Show both.

ANSWER

The group decides what you must prove before a court will accept the record.

PART 3 · WORKED EXAMPLE

An Invoice with No Author

Illustrative: an invoice file recovered from free space

THE CASE

A fake invoice is recovered from free space on a shared accounts PC. Its metadata is gone.

QUESTION

Which clues could link it to one person?

1	**Logged on with the suspect's password** Circumstantial evidence	LINKS
2	**A colleague saw the suspect at the PC** A witness	LINKS
3	**Three clerks share the PC** Others could have made it	WEAKENS
4	**The file names a real supplier** Says nothing about the author	NEITHER

ANSWER

Only clues around the file can name its author. Record any that weaken the link too.

LESSON 2.1 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

What digital evidence is

Anything stored or sent digitally, found in many devices; US courts treat it as physical.

Rules of evidence

Hearsay, business records, and originals or correct copies decide what is accepted.

Financial cases

Accounting records are business records; authorship needs clues around the file.

Handle every record as if you will have to prove how it was made and kept.

UP NEXT

Preparing and Securing the Scene

Next: planning a search, and keeping a scene unchanged until it is recorded.

- **Private-sector scenes**
- **Preparing for a search**
- **Securing the scene**

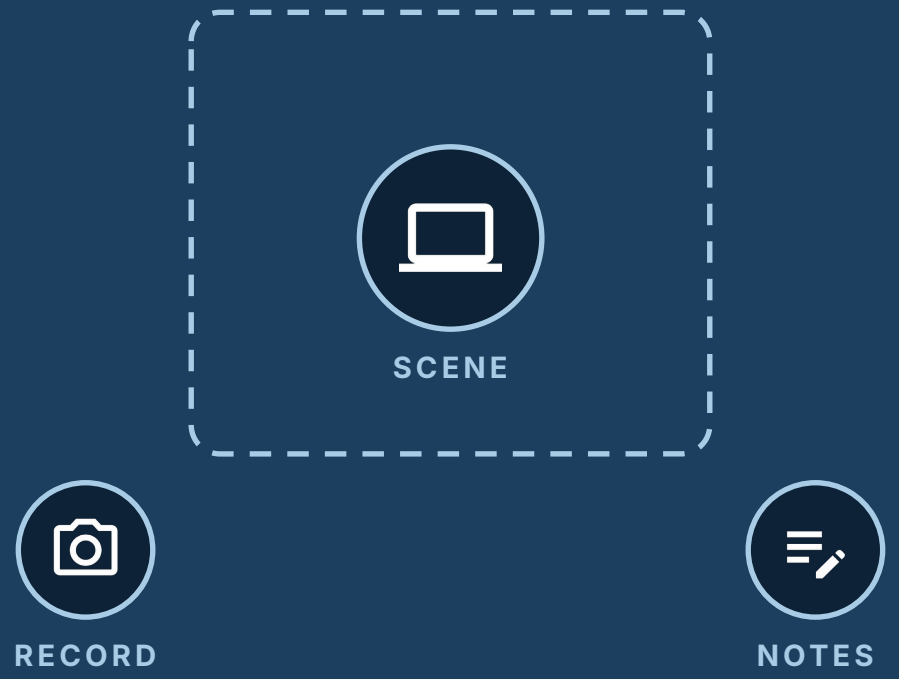
DFBOOK DIGITAL FORENSICS · LESSON 2.2

Preparing and Securing the Scene

Plan the search, protect the scene, decide on running devices

IN THIS LESSON

1. Private-sector scenes
2. Preparing for a search
3. Securing the scene



LESSON 2.2 · OVERVIEW

Preparing and Securing the Scene

Digital evidence is volatile, so a slow or careless start can lose it. Good preparation, a secure perimeter and care with running devices prevent that.

BY THE END YOU WILL BE ABLE TO

1. Handle a crime found in a company case
2. Prepare for a search
3. Secure and record a scene
4. Decide what to do with a running computer

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Case study: the fingerprint that was kept for years



A print found

Kept by the detective as the killer's

A new database

Years later, the print is checked

The match

It was the detective's own print

Professional curiosity: the urge of officers and other professionals to look around a scene. It is a common cause of lost or contaminated evidence.

Digital evidence is just as easily spoiled: even starting a computer changes data such as file dates.

INTRODUCTION · ROADMAP

Your Route Through Lesson 2.2

Each part answers one question, in order

1

PART 1

Private-sector scenes

What changes when the scene is a workplace?

2

PART 2

Preparing for a search

What must be decided before going to the scene?

3

PART 3

Securing the scene

How do you keep the scene and its devices unchanged?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Private-Sector Scenes

Workplace scenes, and what to do if a crime turns up

01

IN THIS PART YOU WILL LEARN TO

- Compare company and police scenes
- Handle a crime found in a company case

KEY TERMS

Reasonable suspicion **Commingled data**

PART 1 · COMPARISON

Company and Police Scenes

The same evidence, under very different control

A COMPANY SCENE

Controlled by management

- A contained office or work area
- Inventory records show the hardware and software
- A banner and policy may allow inspection without a warrant

A POLICE SCENE

Not controlled

- The devices and their use may be unknown
- A warrant is needed to search and seize
- Many officers and specialists may be present

Reasonable suspicion: most company policies require a sound reason to believe a rule is being broken before an inquiry starts.

PART 1 · METHOD

When a Company Inquiry Finds a Crime

Four steps, in order, with the company's lawyer

1. **Check it is a crime**
   Decide with the company's lawyer whether it meets the criteria.
   — WITH THE LAWYER

2. **Inform management**
   Confidential business data may be mixed in with the evidence.
   — WITH MANAGEMENT

3. **Report it**
   Give the police the evidence, and ask the court to protect company secrets.
   — TO THE POLICE

4. **Stop collecting**
   Only under a court order, or you become an agent of the police.
   — AND THEN WAIT

Commingled data: evidence of a crime mixed with confidential company data, such as design plans.

PART 1 · WORKED EXAMPLE

The Engineer's Folder

Case study: illegal images stored with secret design plans

THE CASE

A policy check finds illegal images of children next to secret bicycle plans in an engineer's folder.

QUESTION

What should the examiner do?

1. **Stop examining at once** — Do not open, copy or keep the images — FIRST
2. **Report to the police without delay** — With the lawyer, and a sworn statement — REQUIRED
3. **Ask the court to protect the plans** — So they do not become public record — PROTECT
4. **Wait for a court order** — Before collecting anything more — THEN WAIT

ANSWER

Report the crime, protect the company's secrets, and stop until a court order arrives.

PART 1 · CONTRAST

Compliance Investigations

Audits that check staff follow policy, contracts and regulations

OVERT

Open inspections

- Routine audits of operations and resources
- Workflow reviews to find improvements

COVERT

Hidden monitoring

- Network sniffers that capture traffic
- Keyloggers that record keystrokes and screens

Before covert monitoring: it must be directed by management and set out in a policy telling staff they have no privacy on company systems. Without one, the company can be sued. Check local privacy law.

PART 2 OF 3

Preparing for a Search

What must be settled before anyone goes to the scene

02

IN THIS PART YOU WILL LEARN TO

- List what to find out before a search
- Choose the right field kit

KEY TERMS

Field kit **Hazmat** **Scene leader**

PART 2 · METHOD

Questions Before a Search

Seven things to find out, from the complainant, an informant or a manager

1. **What is the case?**
   Private or public, and how serious.
2. **Which devices and systems?**
   Computers, phones, and any cloud storage.
3. **Can you seize them?**
   Or copy the data on site?
4. **What is the location like?**
   Hazards, such as hazardous materials (hazmat).
5. **Who is in charge?**
   One leader for a large scene.
6. **What expertise is needed?**
   Servers, databases or mobile devices.
7. **Which tools?**
   An initial or an extensive field kit.

Cloud data may need its own warrant or court order.

Move quickly: keep suspects away, and block phones from wireless networks.

PART 2 · ROUTES

Seizing or Copying on Site

Who decides, the usual outcome, and what to watch for

	WHO DECIDES	USUAL OUTCOME	WATCH FOR
Police case	A warrant sets what may be taken	**Seize whole computers and media**	Tell the judge if seizure would ruin a business
Company case	Company policy and management	**Often only a copy of the drive**	Large drives take hours to copy

In Malaysia, Criminal Procedure Code section 116B covers access to computerised data in a search, including passwords. Confirm with a legal adviser.

PART 2 · CATEGORIES

Field Kits

Two kits, compared on the same three points

Initial-response kit

PURPOSE
Light and quick: copy data, then return

EXAMPLES
Laptop, boot media, large drive, forms, bags

ALSO
Toolkit, wrist bands, camera, labels, torch

Extensive-response kit

PURPOSE
The initial kit, plus all you can afford

EXAMPLES
Manuals, power strips, gloves, boxes, spare drives

ALSO
Bolt cutters, hand truck, magnifier, tape measure

At the scene, take out only what you need: it protects the kit and means fewer items to track.

PART 3 OF 3

Securing the Scene

Keeping the scene, and every device in it, unchanged

03

IN THIS PART YOU WILL LEARN TO

- Secure and record a scene
- Decide what to do with a running computer

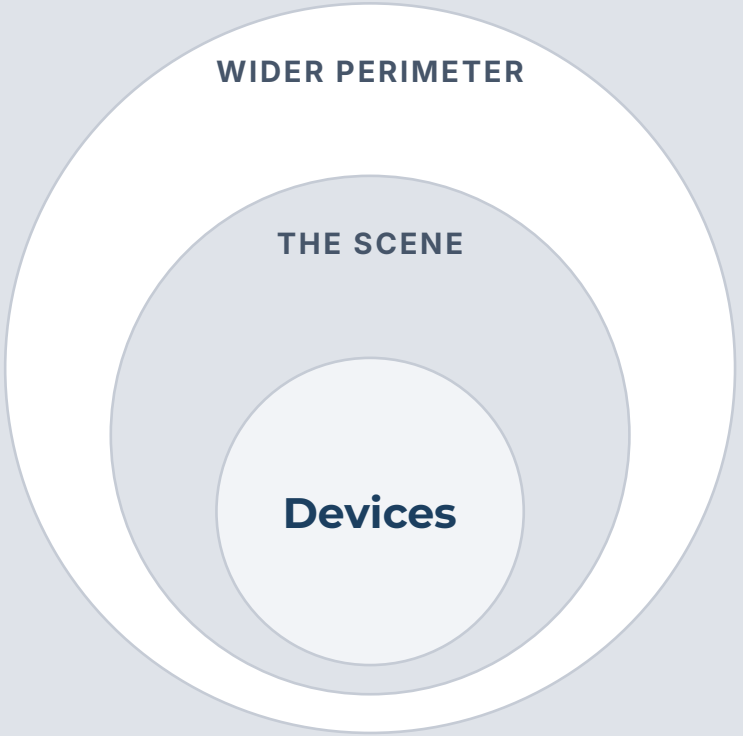
KEY TERMS

Perimeter **Order of volatility** **Orderly shutdown**

PART 3 · STRUCTURE

Securing the Scene

Control a wider area than the scene itself



Restrict access

Barrier tape and guards; only people with a reason may enter. No phone photos.

Go wider, then shrink

Shrinking a perimeter is easier than expanding it.

Say nothing

Be courteous to onlookers; refer the media to a spokesperson.

Computers may also hold fingerprints or DNA on keyboards.

PART 3 · SEQUENCE

Recording the Scene

Three records, made before anything is moved

STEP 1

Start a journal

Arrival time, people met, and every task. It becomes your examiner notes.

STEP 2

Photograph and film

Wide, then close-ups: computer backs, keyloggers, under desks, ceilings, doors. Pan slowly; log shots.

STEP 3

Sketch the room

Rough plan with distances from fixed points, redrawn neatly for the report.

Label every cable

Before photographing the back, so the system can be rebuilt in the lab.

Look for written clues

Passwords, PINs and bank account numbers, in drawers, bins or notes.

PART 3 · DECISION

A Computer Found on or Off

Recommended practice, as two questions

1 Is it switched off?

YES → **Leave it off.** First check it is not just asleep.

NO ↓

2 Encrypted (locked with a key), or mid-task?

YES → **Keep it powered.** Get a specialist; special kits keep power on.

NO ↓

OTHERWISE · FOR A RUNNING COMPUTER

1. Photograph each screen; write down each window
2. Save open work to external media
3. Close the programs
4. Shut down the system's normal way

⚠ Touch as little as possible: no browsing or keys; never save to its drive; agree the shutdown with your supervisor; log every action.

PART 3 · CONCEPT

Volatility and Hidden Traps

What is lost first, and what a suspect may have set up

RAM (working memory) → **Files saved on the drive**

LOST IN MILLISECONDS — CAN LAST FOR YEARS

Order of volatility: how long each kind of information lasts on a system; capture the shortest-lived first. Logs may lose records if power is simply cut.

TRAPS TO WATCH FOR

Altered commands

A familiar command changed to delete everything.

Wipe on wrong password

A screensaver set to erase the drive.

PART 3 · METHOD

Completing the Scene

Four rules for collecting, and what to take away

1. **One collector**
   If possible, one person collects and logs all evidence.
2. **Tag everything**
   Date and time, serial number, make and model, and collector.
3. **Two logs**
   Keep two separate logs, then reconcile them.
4. **Constant control**
   Never leave the evidence or the scene unguarded.

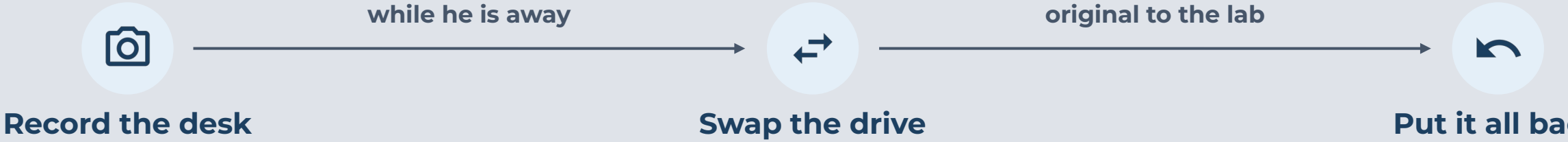
TAKE AWAY

- **Hardware and devices**
- **Software**
- **All media**
- **Manuals, printouts, notes**

PART 3 · WORKED EXAMPLE

The Book at Edward's Desk

Case study: copying a drive without the employee knowing



1 What is the concern?

Edward is suspected of writing a book on his company computer.

2 How is it kept covert?

Photograph the desk, measure the chair, and note where items sit.

ANSWER

A working copy keeps the desk running; the original goes to the lab, and the desk looks untouched.

LESSON 2.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Private-sector scenes

Policy gives access. If a crime turns up, check with the lawyer, report it, then stop.

2

Preparing for a search

Seven questions, from the type of case to the field kit, answered before you go.

3

Securing the scene

Wide perimeter, journal, photos and sketch; then decide carefully about running devices.

Record everything before you touch anything.

UP NEXT

Bagging, Tagging and the Chain of Custody

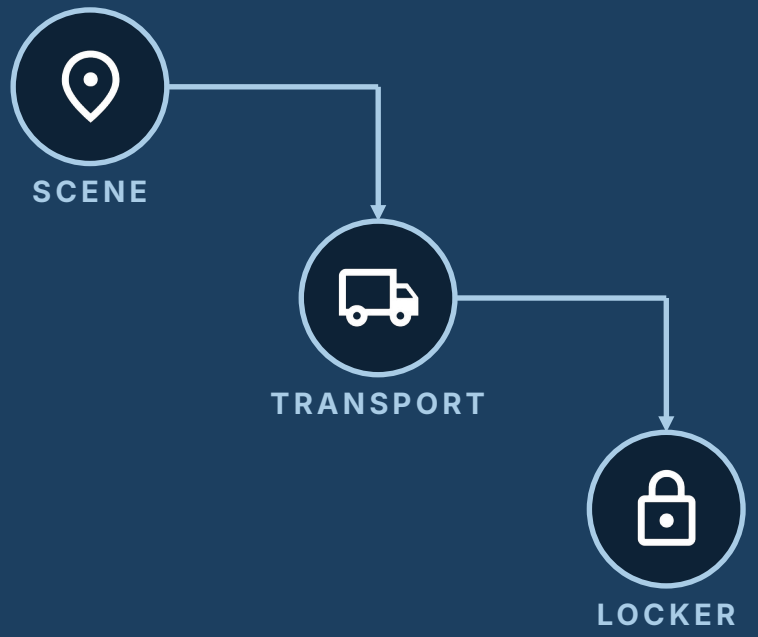
Next: labelling, sealing, moving and storing evidence, and recording every handover.

- **The chain of custody**
- **Bagging and tagging**
- **Storing and keeping evidence**

DFBOOK DIGITAL FORENSICS · LESSON 2.3

Bagging, Tagging and the Chain of Custody

Label, seal, move and store evidence, and record every handover



IN THIS LESSON

1. The chain of custody
2. Bagging and tagging
3. Storing and keeping evidence

LESSON 2.3 · OVERVIEW

Bagging, Tagging and Custody

Evidence is only as good as the record of who held it. This lesson covers the forms, labels, seals and storage that keep that record unbroken.

BY THE END YOU WILL BE ABLE TO

1. Explain the chain of custody
2. Complete an evidence custody form
3. Bag, tag and seal evidence correctly
4. Transport and store evidence safely

INTRODUCTION · WHY THIS MATTERS

Why This Matters

One gap in the record can undo a whole case

> **Many cases are thrown out because the chain of custody cannot be proved, or has been broken.**
>
> Why the chain of custody matters

Every handover

Who passed the evidence to whom, and when.

Every item

Described well enough to tell it from any other.

Every seal

Tape and initials that show nothing was opened.

Chain of custody: the route evidence takes from the time it is found until the case is closed or goes to court, with every step recorded.

INTRODUCTION · ROADMAP

Your Route Through Lesson 2.3

Each part answers one question, in order

1

PART 1

The chain of custody

How is every handover recorded?

2

PART 2

Bagging and tagging

How is evidence labelled, sealed and moved?

3

PART 3

Storing and keeping evidence

Where is evidence kept, and for how long?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

The Chain of Custody

Recording every person and every handover

01

IN THIS PART YOU WILL LEARN TO

- Trace evidence from scene to court
- Complete an evidence custody form

KEY TERMS

Custody form **Single-evidence** **Multi-evidence**

PART 1 · ROUTE

The Route Evidence Takes

Five stops, and a record at every handover



EACH RECORD SHOWS

- **Who**
- **What**
- **When**
- **Where**

⚠ A gap at any stop, such as an unsigned return, means no one can prove the evidence was not changed.

PART 1 · CATEGORIES

Evidence Custody Forms

What every form does, and the two kinds to use

EVERY FORM ✓ Identifies the evidence ✓ Shows who handled it ✓ Lists the dates and times

Single-evidence form

One item per page

More room to describe the item and what was done to it.

KEEP IT **Travels with the item**

Multi-evidence form

Up to ten items per page

Lists several items from one location, numbered page 1 of 2, and so on.

KEEP IT **Stays in the report file**

Using both gives a second record to cross-check, which is a simple quality control.

PART 1 · STRUCTURE

What a Custody Form Records

Twelve fields, from the case number to the page count

Case number
Given when the case opens

Organisation
Who is investigating

Investigator
The lead, if there are several

Nature of case
A short description

Location
Exactly where it was found

Description
One USB drive, 8 GB

Make, model, serial
Who made it, and its numbers

Recovered by
Starts the chain of custody

Date and time
When custody began

Placed in locker
Which one, and when

Processed by
Each later handling, and why

Page
For example, page 1 of 2

Write the instructions down too, so everyone fills in the form the same way.

PART 1 · WORKED EXAMPLE

A Completed Single-Evidence Form

The Montgomery case: the USB drive from the IT department

SINGLE-EVIDENCE FORM

CASE NO.	Montgomery_72022
DESCRIPTION	One USB drive, 8 GB
MAKE	SanDisk
RECOVERED BY	Examiner's name
DATE AND TIME	20 July 2022, 10:05
LOCKER	Locker A, 11:30

MOCK · ILLUSTRATIVE

1. **Identifies the item**
   Description and make tell this drive from any other.
2. **Starts the chain**
   The name here takes responsibility for the drive.
3. **Fixes the time**
   Custody begins at this date and time.
4. **Shows where it went**
   Which locker, and when it went in.

PART 2 OF 3

Bagging and Tagging

Labelling, sealing and moving evidence safely

02

IN THIS PART YOU WILL LEARN TO

- Tag and bag evidence correctly
- Seal and transport larger items

KEY TERMS

Antistatic bag **Evidence tape** **Tag**

PART 2 · METHOD

Tagging and Bagging

Four rules for small items such as drives and phones

Tag every item
Date and time, serial number, make and model, and the collector's name.

IDENTIFY

Write on the bag first
Fill in the bag's label before the item goes in.

RECORD

Use antistatic bags
Static can destroy data, and not every evidence bag is antistatic.

PROTECT

Label the bag, not the item
If a label could damage the item, put it on the bag instead.

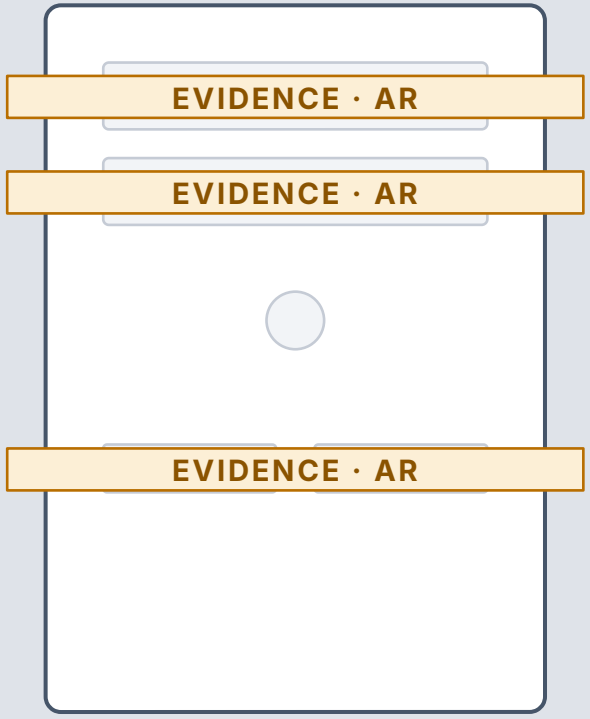
PRESERVE

Identical items: tag each of several identical USB drives or discs, so the form can tell them apart.

PART 2 · EXHIBIT

Sealing a Computer

Evidence tape over every opening, with your initials across it



1. **Tape every opening**
   Drive bays, the power socket, USB slots and any other way in.

2. **Initial the tape first**
   Write your initials across it before applying it.

3. **Why it works**
   New tape would lack your initials, so tampering shows.

4. **Pad and protect**
   Padded containers; antistatic pads and wrist straps when handling.

PART 2 · PROCEDURE

Transporting Evidence

What to pack, and the rules on the way to the lab

PACK

Have on hand

- ✓ Antistatic bags in several sizes
- ✓ Bubble wrap and padded boxes
- ✓ Packing tape and evidence tape
- ✓ Labels and permanent markers

MOVE

On the way

- ✓ Under watch at all times, by you or the custodian
- ✓ No one else may have access
- ✓ Away from heat, cold and magnetic fields
- ✓ Not on a heated car seat or a two-way radio

Evidence custodian: the person assigned to keep the evidence secure, including in transit.

PART 3 OF 3

Storing and Keeping Evidence

The lab, long-term storage, and how long to keep it

03

IN THIS PART YOU WILL LEARN TO

- Store evidence securely in the lab
- Choose media for long-term storage

KEY TERMS

Evidence locker **Retention** **Contraband**

PART 3 · SEQUENCE

Evidence in the Lab

Every movement in and out of the locker is written down

STEP 1

Lock it away

A locked, fireproof locker or cabinet that only authorised people can open.

STEP 2

Sign it out

Record who takes it, when and why. Work on copies, never the original.

STEP 3

Sign it back in

Record its return to the locker, with the date and time.

Copy the form

Keep the form with the evidence, and a photocopy with your examiner notes.

Coming in Lesson 3

Hash values that prove a copy matches the original are added to the form.

PART 3 · COMPARISON

Media for Long-Term Storage

Hard drives and SSDs fail over time, so evidence is archived to other media

Media	Typical life	Capacity	Trade-off
Optical (CD, DVD)	5 to 10 years	700 MB to 4.7 GB	Cheap, but many discs
Tape (DAT, DLT, Super DLT)	DLT: 50 years or more	35 GB (DLT) to 600 GB	Low cost per GB, but slow
M-Disc	1,000 years (maker's claim)	4.7 to 100 GB	For very long storage

Keep two copies of every forensic image, made with different tools if possible. Check restored tape data before relying on it.

PART 3 · CONTRAST

Keeping Evidence, and One Exception

How long evidence is kept, and the one kind no company may keep

RETENTION

How long to keep it

- Police cases: possibly indefinitely; check with the prosecutor or the law
- Company cases: the legal department sets the rules

THE EXCEPTION

Illegal images of children

- Stop at once; do not examine the files
- Call law enforcement; only they may hold it

Contraband: material that is illegal to possess. Only law enforcement may store it.

PART 3 · WORKED EXAMPLE

Finding the Gap in a Custody Log

Illustrative log for one USB drive: where does the chain break?

#	Date and time	Action	Signed by
1	20 Jul, 10:05	Collected from the IT manager	A. Rahman
2	20 Jul, 10:40	Transported to the lab	A. Rahman
3	20 Jul, 11:30	Placed in locker A	A. Rahman
4	**22 Jul, 09:00**	**Signed out for copying**	**S. Lee**
5	**23 Jul, 17:00**	**Returned to locker A**	**S. Lee**



Who held it at each step?



ANSWER

Between entries 4 and 5 the drive was out for 32 hours with no record of where it was kept.

LESSON 2.3 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

The chain of custody

Every handover recorded on single- and multi-evidence forms, from scene to court.

2

Bagging and tagging

Tag, label the bag, use antistatic bags, seal with initialled tape, guard in transit.

3

Storing evidence

Locked locker, signed in and out; two archive copies; contraband goes to the police.

If it is not written down, it did not happen.

UP NEXT

Lesson 2 Summary

Next, a short review brings Lessons 2.1 to 2.3 together before Lesson 3.



Identifying digital evidence



Preparing and securing the scene



Bagging, tagging and custody

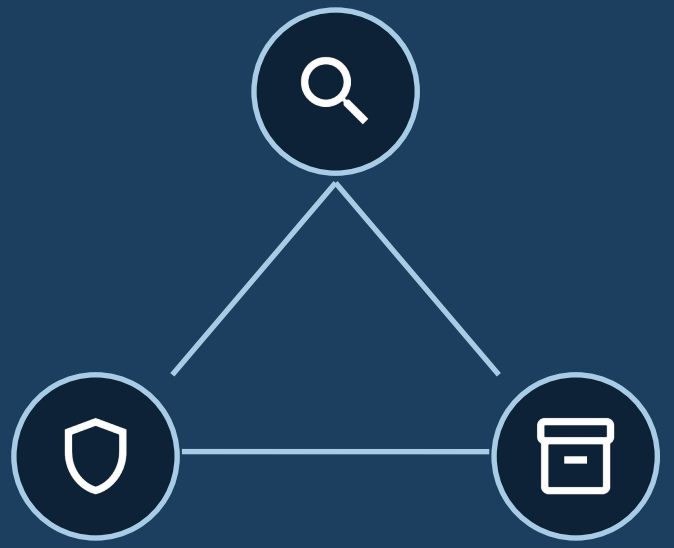
DFBOOK DIGITAL FORENSICS · LESSON 2 · SUMMARY

Evidence at the Scene: Lesson Review

Identify, secure and record, in one short review

IN THIS LESSON

1. Identifying digital evidence
2. Preparing and securing the scene
3. Bagging, tagging and custody



IDENTIFY · SECURE · RECORD

LESSON 2 · REVIEW

Lesson 2 at a Glance

One column for each sub-lesson

2.1 Identifying evidence

1. Anything stored or sent digitally; US courts treat it as physical evidence.
2. Generated records need a working program; stored records need an author.
3. In Malaysia, see the Evidence Act 1950.

2.2 Securing the scene

1. If a company inquiry finds a crime: lawyer, report, then stop.
2. Answer seven questions before a search.
3. Record before touching; if a computer is off, leave it off.

2.3 Bagging and custody

1. Every handover recorded on a custody form.
2. Tag items, use antistatic bags, initial the tape.
3. Locked locker; two archive copies.

LESSON 2 · REVIEW

From Scene to Locker

How the three sub-lessons fit together



Find it, protect it, and write down every step it takes.

UP NEXT

Making and Verifying a Forensic Copy

Lesson 3 shows how examiners copy evidence exactly, and prove the copy is exact.

- **Forensic copies**
- **Hash values**
- **Checking the copy**

